促銷策略

- 心理層面與廣告的影響
- 促銷策略介紹
- 關聯規則與產品推薦
- 實務運用與影片分享

促銷策略與關連規則

一碗拉麵6000元消費者竟然搶著要？從Gogoro、鬍鬚張、一蘭拉麵看定價策略 - 行銷人 (marketersgo.com)

15定价策略，以提高您的销售（与示例） (thatsmyhorse.com)

定價策略》安卓手機，為何「特別容易」被猜出定價？－網民肥皂箱｜商周 (businessweekly.com.tw)

星巴克為什麼要賣月餅？行銷4P：以對的時間、定價，把對的產品放在對的位置－管理知識內參｜商周 (businessweekly.com.tw)

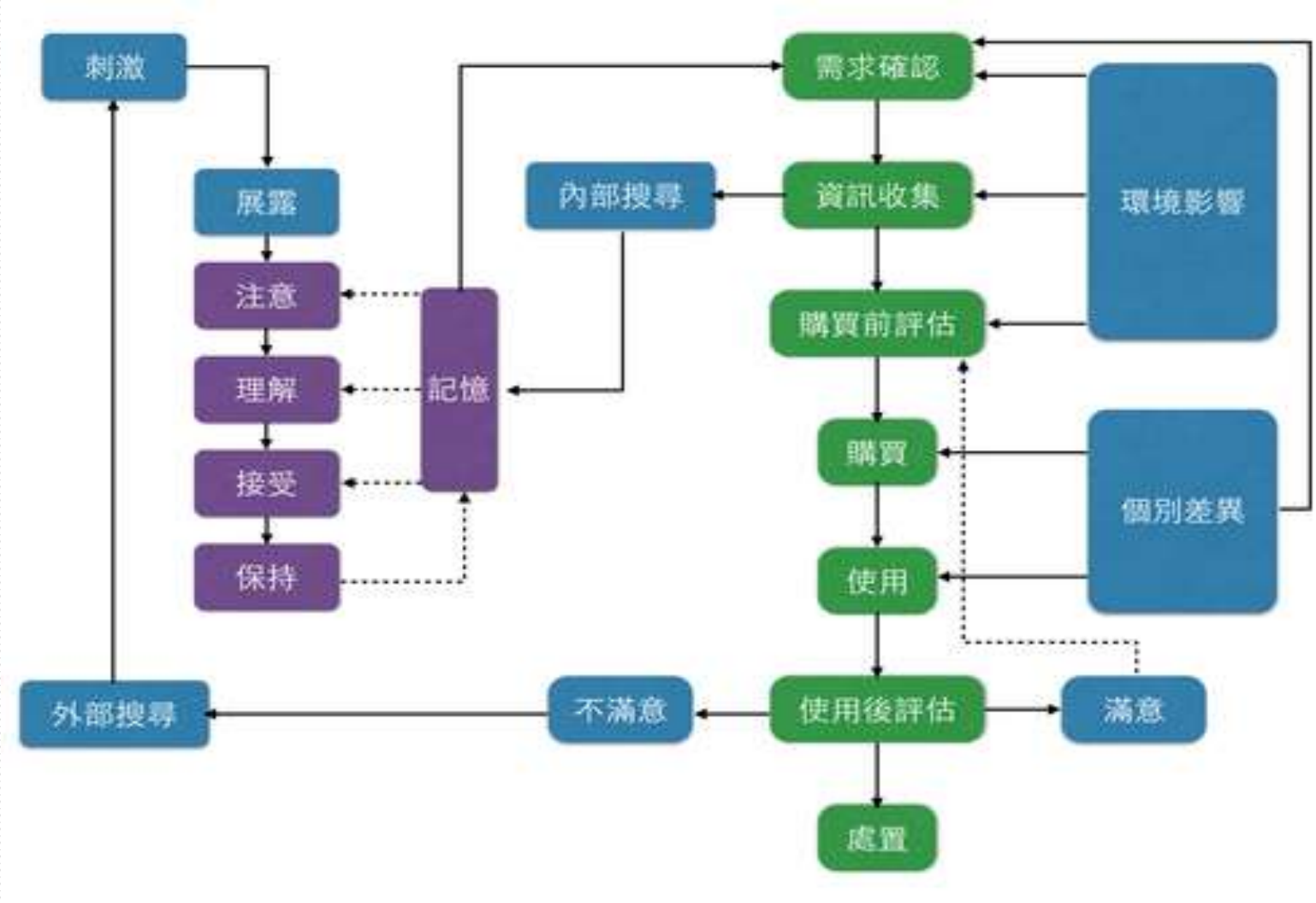
經營網店如何做好商品定價 (訂價)？ - TechTeller (科技說)

有效提升廣告轉換率！吸引消費者注意力，用這6種廣告標題法 - 行銷人 (marketersgo.com)

購買決策模式


Awareness
知曉
Interest
興趣
Search
搜尋
Consideration
考慮
Action
購買行動
Advocacy
擁護
忠誠度
接觸點
店面招牌
展覽活動
公關活動
業務推廣
親友分享
網紅、名人代言
部落格
論壇
社群平台
廣告：看板、傳單、
電視、廣播、網路
內容經營：
網頁
部落格
論壇
社群平台
比較資訊：
價格
功能
外觀
實用性
使用體驗
實體店面
電商平台
消費體驗、感謝卡、
EDM、優惠資訊
商家評論、留言分享
網路分享：
Facebook、YouTube
PTT、部落格
客戶開發

購買資訊處理


刺激
展露
注意
理解
接受
保持
記憶
內部搜尋
需求確認
資訊收集
購買前評估
購買
使用
使用後評估
環境影響
個別差異
不滿意
外部搜尋
滿意
處置

衝動性購買



總是不小心「手滑」多買？簡單抵抗衝動性購物的十大方法 (greenvines.com.tw)

強迫性購買



愛花錢也是一種精神病！認識「強迫購物症」 - Heho健康

蠻牛廣告的魅力

(462) 蠻牛廣告-電話篇(完整60秒) - YouTube

(462) 蠻牛-倒垃圾篇201112 - YouTube

(462) 保力達蠻牛 腳踏車店篇 - YouTube

(462) 0928 蠻牛臨演篇 網路長版 - YouTube

(462) 保力達蠻牛-駕訓班篇 - YouTube

(462) 蠻牛-開鎖篇60秒完整版 - YouTube

汽車廣告的魅力

(462) THE ALL-NEW FORD FOCUS ACTIVE X林依晨 感動鉅獻｜迷路｜FORD TAIWAN - YouTube

(462) V60 終於自我 | 睽違 19 年 陳柏霖與桂綸鎂攜手獻映 100 秒完整版 - YouTube

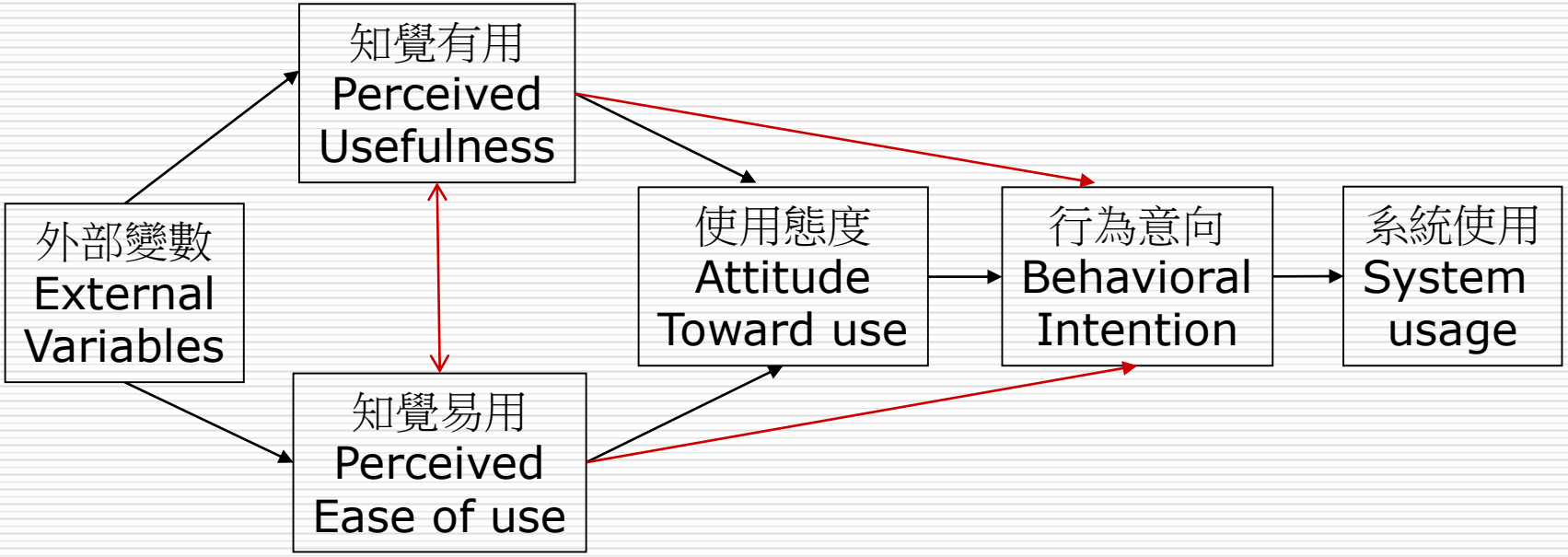
(462) NISSAN KICKS 都會時尚跨界休旅 全新上市 HD - YouTube

(462) [My Baby]三菱 Outlander 2010廣告(完整版)~ Stanley 的第5支廣告喔^^ - YouTube

(462) Ford 2008 Mondeo超溫馨廣告 家庭篇 - YouTube

(462) MONDEO MetroStar CF 盲人賓士篇 - YouTube

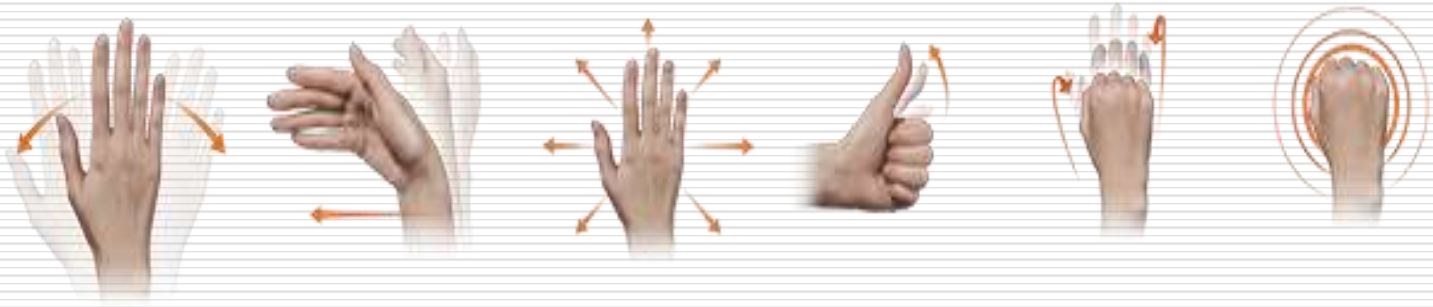
科技接受模式 (TAM)


知覺有用
Perceived
Usefulness
外部變數
External
Variables
知覺易用
Perceived
Ease of use
使用態度
Attitude
Toward use
行為意向
Behavioral
Intention
系統使用
System
usage

智能電視 (Smart TV)

Typical smart-TV brands	Samsung	Panasonic	LG	Sony
Engineering features (EFs): *PU*				
E1- On-demand video service	*	*	*	*
E2- Game consoler entertainment	*	*	*	*
E3- 2D/3D virtual reality	*	*	*	*
E4- Recording and cloud storage	*	*	*	*
E5- Searching & recommendation	*	*	*	*
E6- Gesture recognition and control	*		*	
E7- Voice control	*	*	Device	Device
E8- Face recognition	*	*		
E9- Individual APP platform	*	*	*	*
E10- Smart device enabled control	*	*	*	*
Gesture features (GFs): *PEOU*				
G1- Wave a hand	*		*	
G2- Flip to left/right	*		*	
G3- Mouse-like move	*		*	
G4- Thumb up	*			
G5- Grab to select TV channels	*		*	
G6- Long grab adjust volume	*		*	
G7- Rotate and return to previous step	*		*	
G8- Grab and move to power on/off	*			
G9- Wave two hands	*			
G10- Grab to zoom in/out a photo	*			
G11- Grab and rotate a photo	*			

智能電視 (Smart TV)



G1 G2 G3 G4 G5 G6



G7 G8 G9 G10 G11

案例：松下的美顏智慧鏡子

- 松下（Panasonic）開發了的美顏智慧鏡子「Beauty Smart Mirror」，它可以做到化妝預覽，完全運用到擴增實境的技術，讓使用者可以未上妝先體驗上妝後的風格。可以模擬的有眉妝、眼妝、腮紅、唇妝和鬍子等，選擇後再看整體的化妝效果。還可以看到幾個使用者用過這個化妝模式以及多少使用者喜歡這個內容的社群激勵部分。

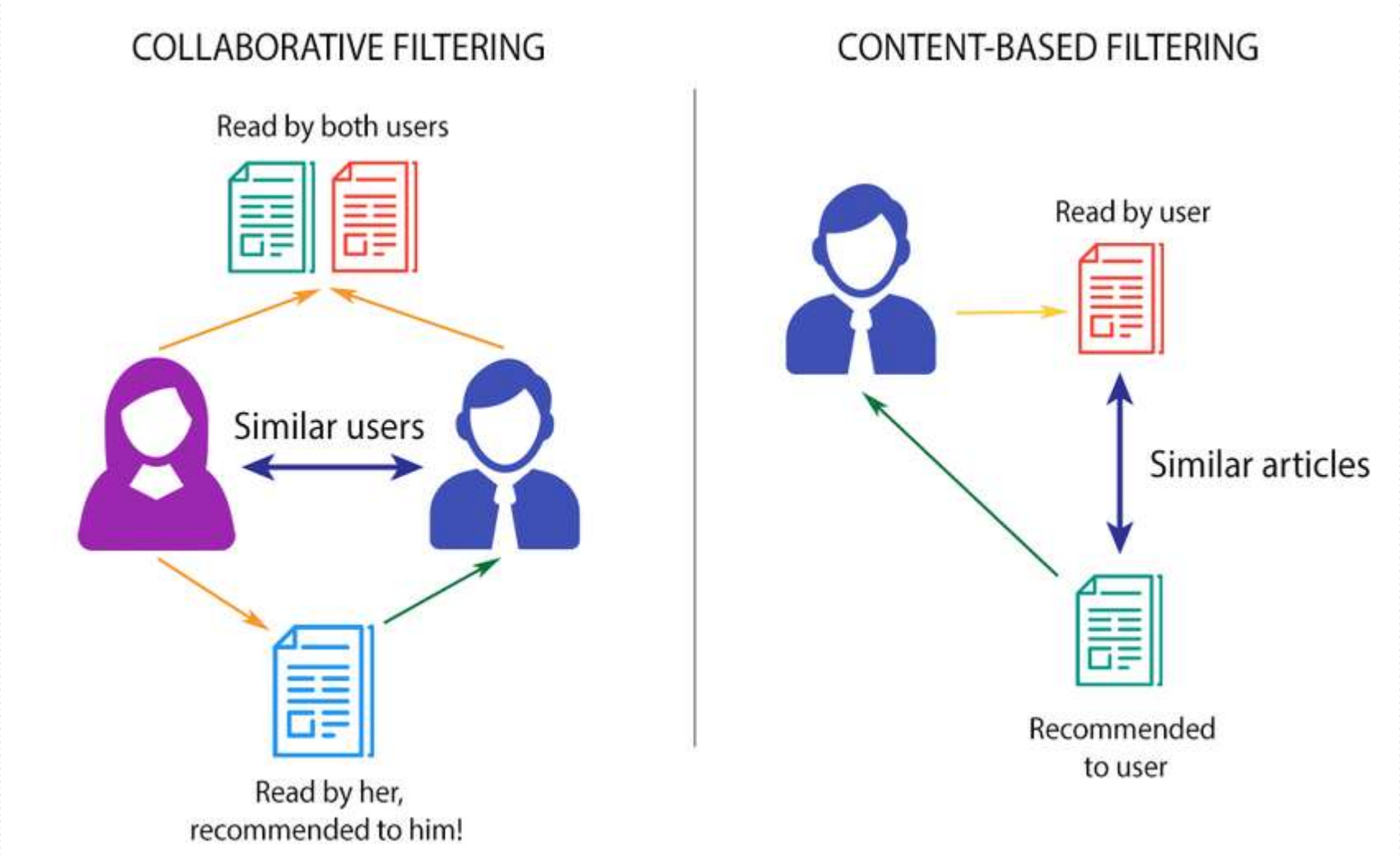


(661) Panasonic's Innovative Future Mirror in Action at Panasonic CES 2016 - YouTube

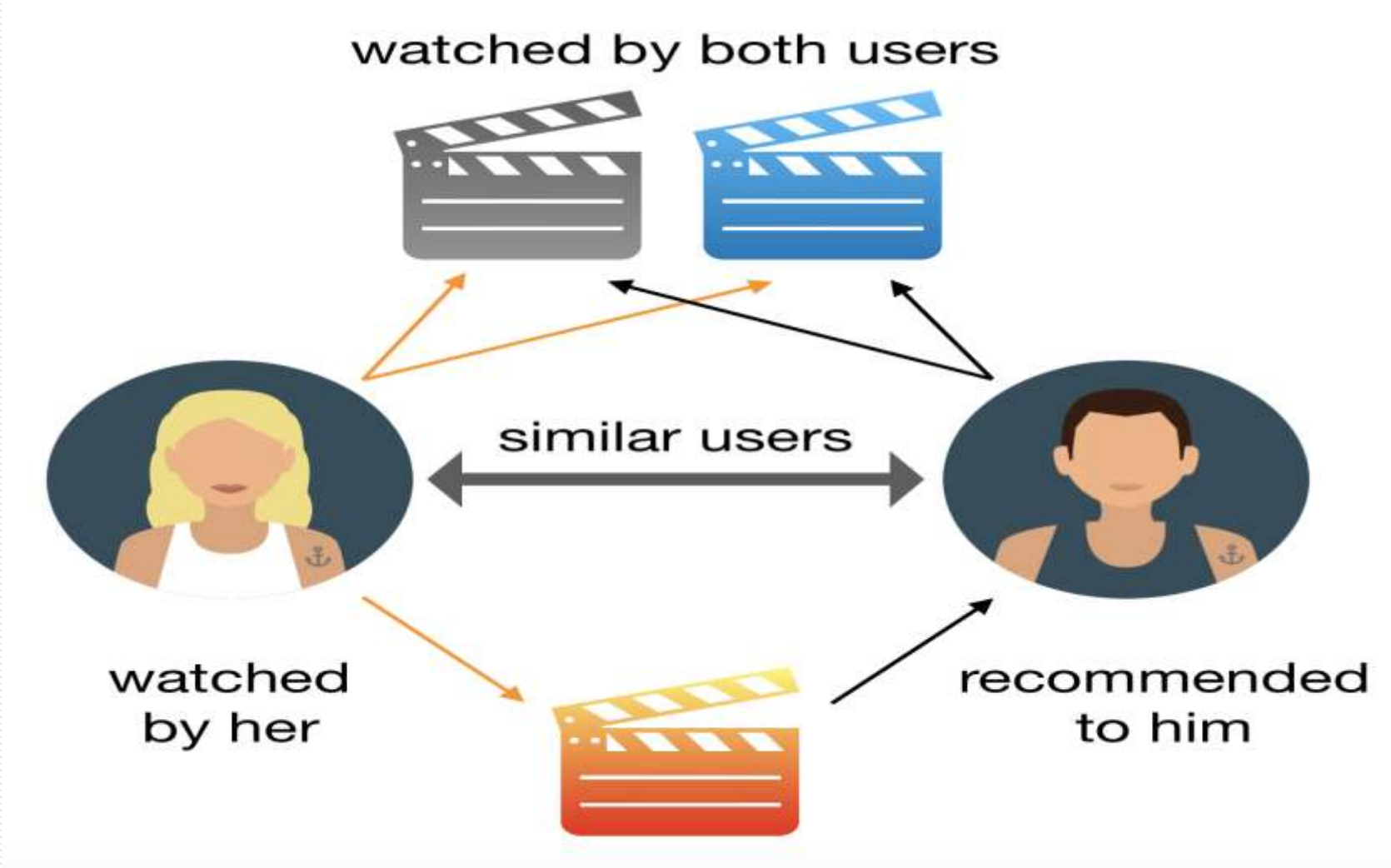
案例：松下的美顏智慧鏡子

- ✓ 使用人工智慧，做到皮膚分析的功能，利用毛孔、斑點、皮膚透亮程度、皺紋及微笑紋五項內容做成雷達圖，可以看出你的皮膚是否需要水分，而且跟之前的紀錄比較，看皮膚狀況是否改進，如果這是使用保養品的結果的話，就可以知道保養品的使用是否有效，並具體提出改善的方式，像是遮瑕膏等。
- ✓ 同時它還能跟智慧型手機上的行事曆同步，提醒你是要上班、出席聚會，對應到不同的妝。在鏡子的底部具備化妝面膜製作機，經過演算法分析後生成一種化妝面膜，用戶透過設備將妝敷上臉部，就完成化妝了。
- ✓ 不過2016年底版本的設備產生的這個面膜利用數分鐘製作好，但是卻需一天風乾後才能使用，並不符合消費者需求。松下智慧鏡子已經研發多年，但現在仍屬實驗性產品。

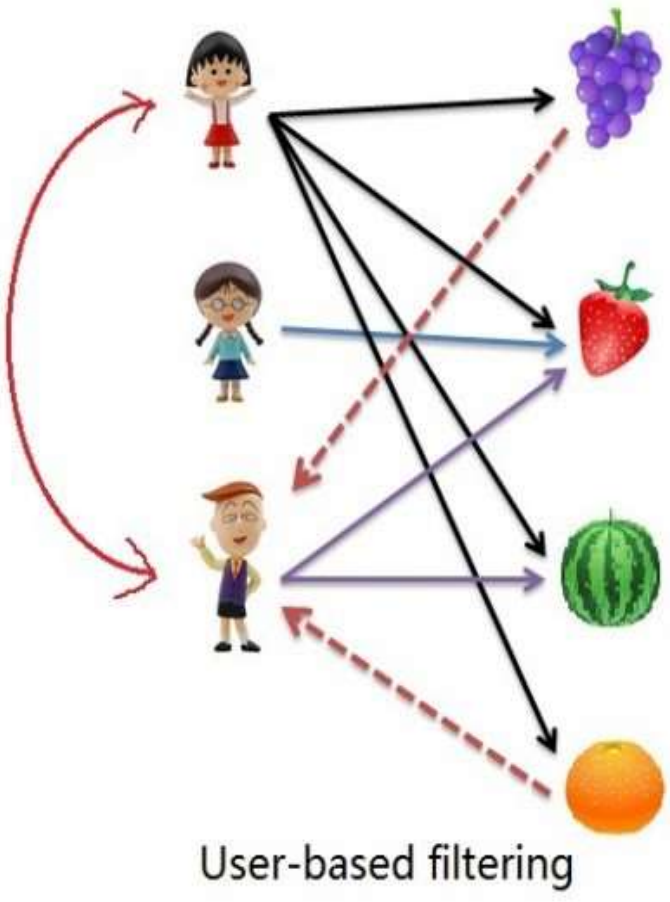
產品推薦模式


COLLABORATIVE FILTERING
Read by both users
Similar users
Read by her,
recommended to him!
CONTENT-BASED FILTERING
Read by user
Similar articles
Recommended
to user

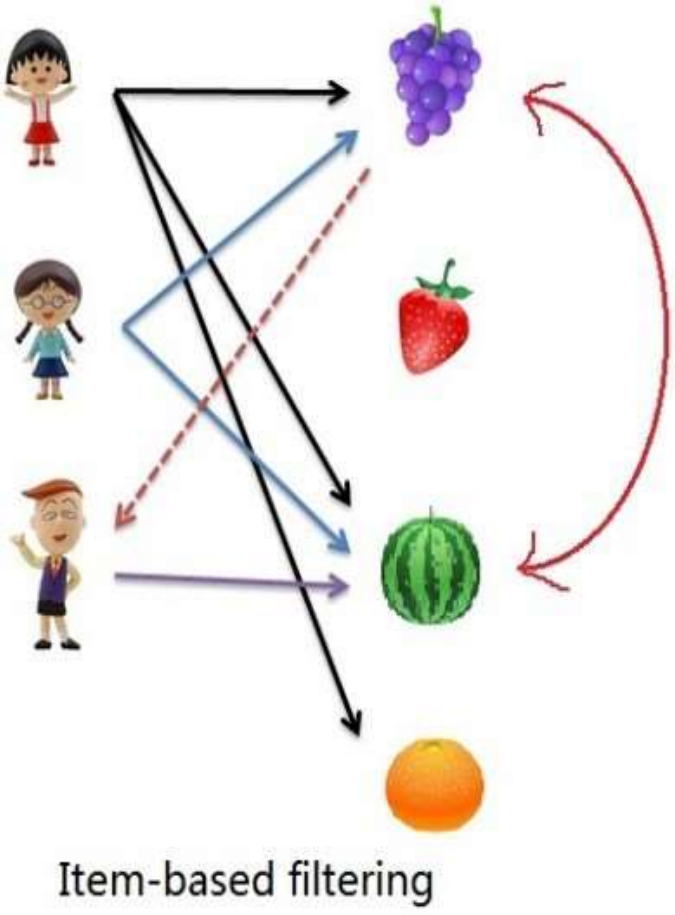
內容導向推薦模式


watched by both users
similar users
watched
by her
recommended
to him

偕同過濾推薦模式



User-based filtering



Item-based filtering

關聯規則 Association Rule

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

(購物籃分析)

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

{Diaper} → {Beer},
{Milk, Bread} → {Eggs,Coke},
{Beer, Bread} → {Milk},

Implication means co-occurrence, not causality!

Definition: Frequent Itemset

- **Itemset**
  - A collection of one or more items
    - Example: {Milk, Bread, Diaper}
  - k-itemset
    - An itemset that contains k items
- **Support count (σ)**
  - Frequency of occurrence of an itemset
  - E.g. σ({Milk, Bread, Diaper}) = 2
- **Support**
  - Fraction of transactions that contain an itemset
  - E.g. s({Milk, Bread, Diaper}) = 2/5
- **Frequent Itemset**
  - An itemset whose support is greater than or equal to a *minsup* threshold

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- Association Rule (LHS → RHS)
  - An implication expression of the form X → Y, where X and Y are itemsets
  - Example:
    {Milk, Diaper} → {Beer}

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- Rule Evaluation Metrics
  - Support (s)
    - Fraction (probability) of transactions that contain both X and Y
  - Confidence (c)
    - Measures (conditional probability) how often items in Y appear in transactions that contain X

Example:

$$\{\text{Milk}, \text{Diaper}\} \Rightarrow \text{Beer}$$

$$s = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{|\text{T}|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\sigma(\text{Milk}, \text{Diaper})} = \frac{2}{3} = 0.67$$

Association Rule Mining Task

- Given a set of transactions T, the goal of association rule mining is to find all rules having
  - support ≥ *minsup* threshold
  - confidence ≥ *minconf* threshold

- Brute-force approach:
  - List all possible association rules
  - Compute the support and confidence for each rule
  - Prune rules that fail the *minsup* and *minconf* thresholds

  ⇒ Computationally prohibitive!

Mining Association Rules

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

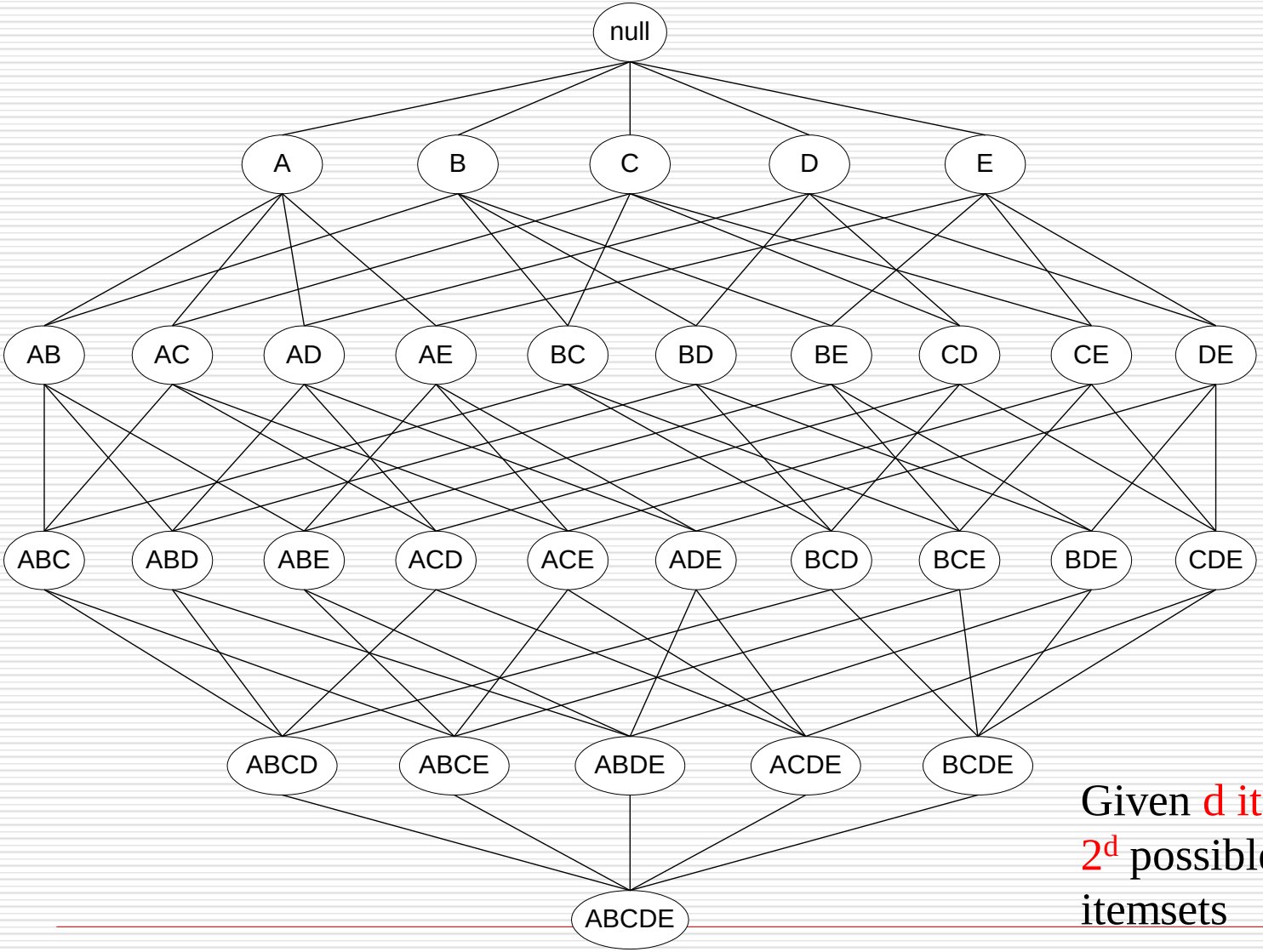
Example of Rules:

{Milk,Diaper} → {Beer} (s=0.4, c=0.67)
{Milk,Beer} → {Diaper} (s=0.4, c=1.0)
{Diaper,Beer} → {Milk} (s=0.4, c=0.67)
{Beer} → {Milk,Diaper} (s=0.4, c=0.67)
{Diaper} → {Milk,Beer} (s=0.4, c=0.5)
{Milk} → {Diaper,Beer} (s=0.4, c=0.5)

Observations:

- All the above rules are binary partitions of the same itemset:
  {Milk, Diaper, Beer}
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, we may decouple the support and confidence requirements

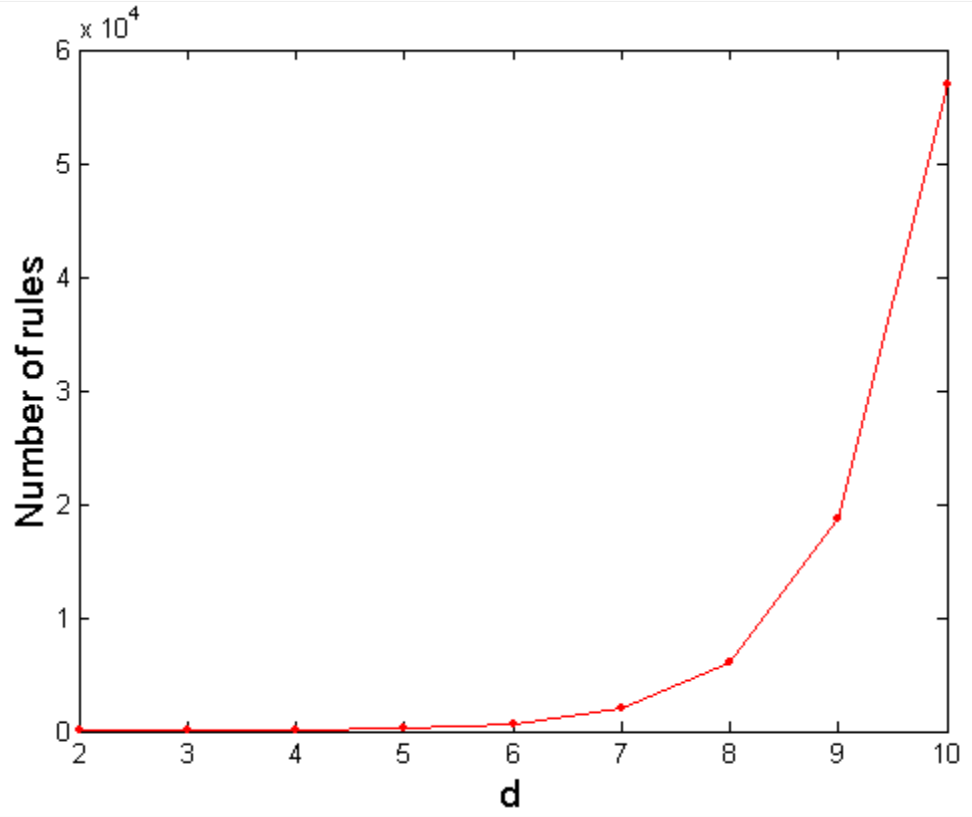
Frequent Itemset Generation



Given d items, there are 2^d possible candidate itemsets

Computational Complexity (二項式定理)

- Given d unique items:
  - Total number of itemsets = 2^d
  - Total number of possible association rules:



$$R = \sum_{k=1}^{d-1}\left[\binom{d}{k} \times \sum_{j=1}^{d-k}\binom{d-k}{j}\right]$$

$$= 3^d - 2^{d+1} + 1$$

If d=6 items, R = 602 rules

Frequent Itemset Generation Strategies

- Reduce the number of candidates (M)
  - Complete search: $M=2^d$
  - Use pruning techniques to reduce M
- Reduce the number of transactions (N)
  - Reduce size of N as the size of itemset increases
  - Used by DHP and vertical-based mining algorithms
- Reduce the number of comparisons (NM)
  - Use efficient data structures to store the candidates or transactions
  - No need to match every candidate against every transaction

Reducing Number of Candidates

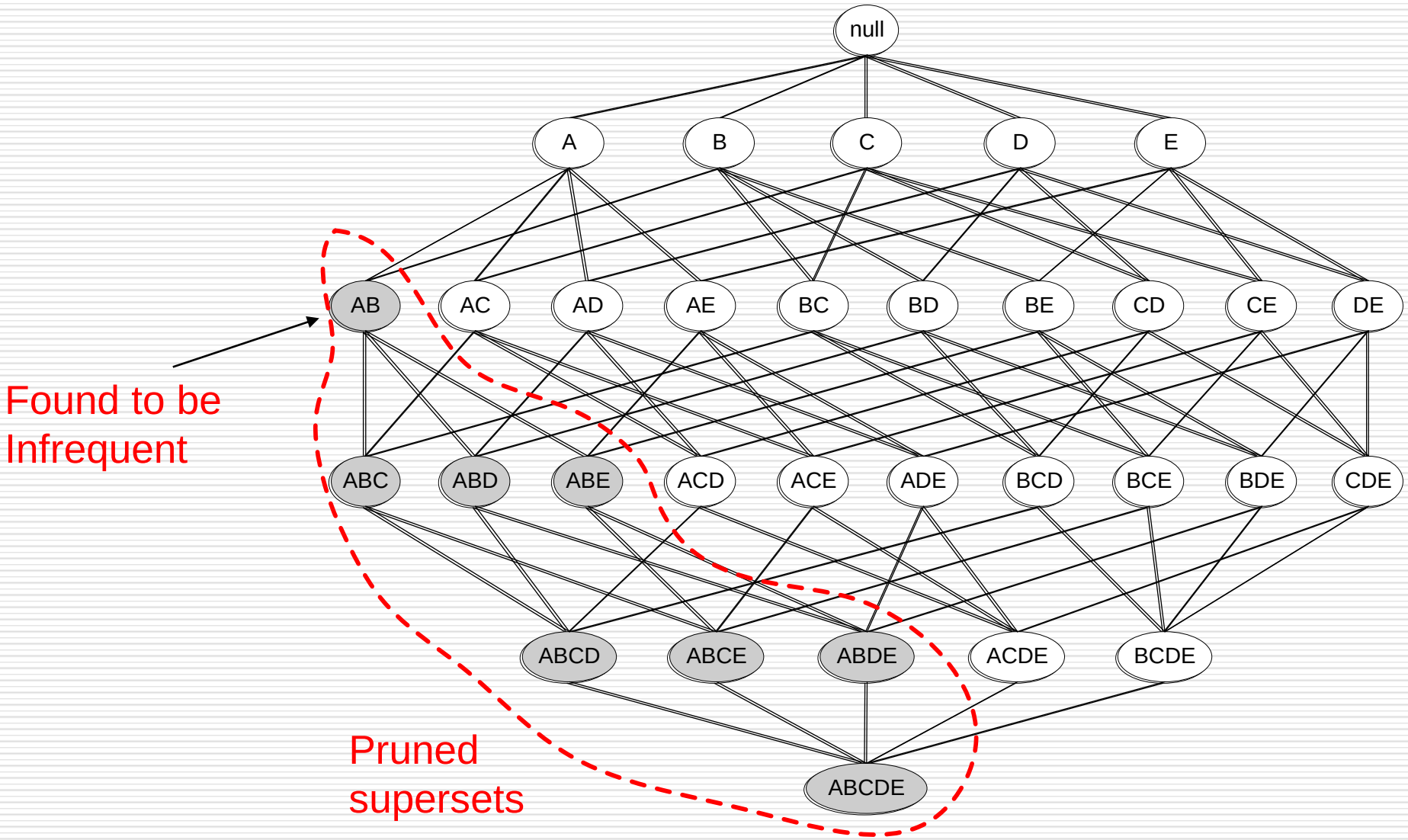
- Apriori principle:
  - If an itemset is frequent, then all of its subsets must also be frequent

- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

  - Support of an itemset never exceeds the support of its subsets
  - This is known as the anti-monotone property of support

Illustrating Apriori Principle



Drawback of Confidence

	Coffee	$\overline{\text{Coffee}}$	
Tea	15	5	20
$\overline{\text{Tea}}$	75	5	80
	90	10	100

Association Rule: Tea $\rightarrow$ Coffee

Confidence= P(Coffee|Tea) = 0.75 (15/20)

but P(Coffee) = 0.9 (90/100)

$\Rightarrow$ Although confidence is high, rule is misleading

$\Rightarrow$ P(Coffee|$\overline{\text{Tea}}$) = 0.9375 (75/80)

Statistical Independence

- Population of 1000 students
  - 600 students know how to swim (S)
  - 700 students know how to bike (B)
  - 420 students know how to swim and bike (S,B)

  - $P(S \wedge B) = 420/1000 = 0.42$
  - $P(S) \times P(B) = 0.6 \times 0.7 = 0.42$

  - $P(S \wedge B) = P(S) \times P(B)$ => Statistically independent
  - $P(S \wedge B) > P(S) \times P(B)$ => Positively correlated
  - $P(S \wedge B) < P(S) \times P(B)$ => Negatively correlated

Statistical-based Measures

- Measures that take into account statistical dependence

Lift is equal to interest

$$Lift = \frac{P(Y \mid X)}{P(Y)}$$

Support: LHS & RHS同時發生的機率
Confidence: LHS發生的前提下，RHS也發生的條件機率
Lift: LHS & RHS同時發生的機率除以LHS & RHS個別發生的機率

$$Interest = \frac{P(X,Y)}{P(X)P(Y)}$$

$$PS = P(X,Y) - P(X)P(Y)$$

$$\phi - coefficient = \frac{P(X,Y) - P(X)P(Y)}{\sqrt{P(X)[1-P(X)]P(Y)[1-P(Y)]}}$$

Example: Lift/Interest

	Coffee	$\overline{\text{Coffee}}$	
Tea	15	5	20
$\overline{\text{Tea}}$	75	5	80
	90	10	100

Association Rule: Tea $\rightarrow$ Coffee

Confidence= P(Coffee|Tea) = 0.75

but P(Coffee) = 0.9

$\Rightarrow$ Lift = 0.75/0.9=0.15/(0.2*0.9)= 0.8333 (< 1, therefore is negatively associated)

預測的種類

- There are four basic types of forecasts.
  - Qualitative (老闆、主管或專家說了算)
  - Time series analysis (根據過去歷史資料)
  - Causal relationships (迴歸就是用 X預測Y)
  - Simulation (需考慮機率或不確定性)

- Time series (時間序列) analysis is based on the idea that data relating to past demand can be used to predict future demand.

績效指標衡量

$$MSE = \frac{\sum(Actual_t - Forecast_t)^2}{n-1}$$ Mean Squared Error

$$MAD = \frac{\sum|Actual_t - Forecast_t|}{n}$$ Mean Absolute Deviation

$$MAPE = \frac{\sum\left|\frac{Actual_t - Forecast_t}{Actual_t}\right|}{n} * 100\%$$ Mean Absolute Percentage Error

Simple Moving Average

- Forecast is the average of a fixed number of past periods (以過去歷史資料的平均預測未來).
- Useful when demand is not growing or declining rapidly and no seasonality is present.
- Removes some of the random fluctuation from the data.
- Selecting the period length is important.
  - Longer periods provide more smoothing.
  - Shorter periods react to trends more quickly.

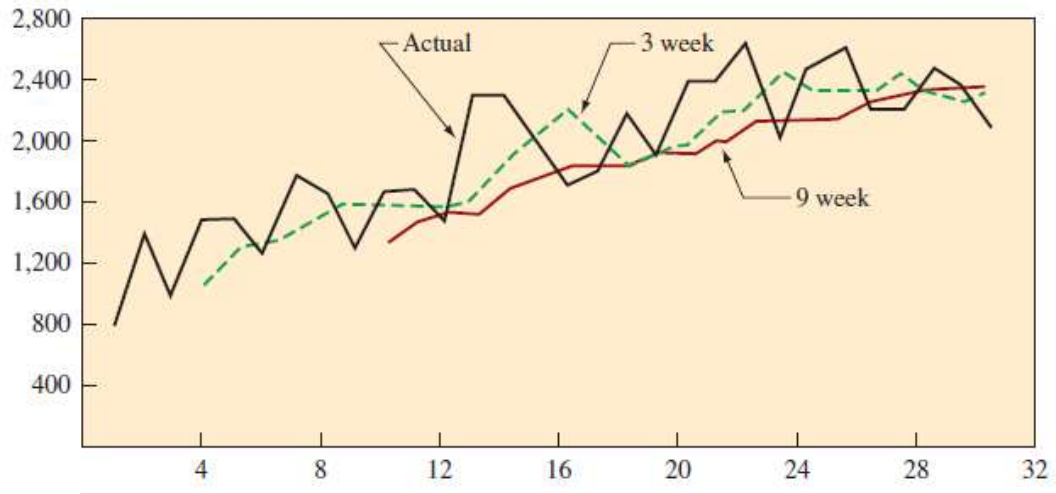
Simple Moving Average Formula

就是股市當中的五日、十日、20日移動平均線

- $F_t = \frac{A_{t-1}+A_{t-2}+A_{t-3}+\cdots+A_{t-n}}{n}$
- F_t - Forecast for period t (the next period)
- n – Number of periods to be averaged
- A_{t-1} - Actual occurrence in the past period
- A_{t-2}, A_{t-3}, and A_{t-n} - Actual occurrences two periods ago, three periods ago, and so on up to n periods ago

Simple Moving Average – Example

Week	Demand	3 Week	9 Week	Week	Demand	3 Week	9 Week
1	800			16	1,700	2,200	1,811
2	1,400			17	1,800	2,000	1,800
3	1,000			18	2,200	1,833	1,811
4	1,500	1,067		19	1,900	1,900	1,911
5	1,500	1,300		20	2,400	1,967	1,933
6	1,300	1,333		21	2,400	2,167	2,011
7	1,800	1,433		22	2,600	2,233	2,111
8	1,700	1,533		23	2,000	2,467	2,144
9	1,300	1,600		24	2,500	2,333	2,111
10	1,700	1,600	1,367	25	2,600	2,367	2,167
11	1,700	1,567	1,467	26	2,200	2,367	2,267
12	1,500	1,567	1,500	27	2,200	2,433	2,311
13	2,300	1,633	1,556	28	2,500	2,333	2,311
14	2,300	1,833	1,644	29	2,400	2,300	2,378
15	2,000	2,033	1,733	30	2,100	2,367	2,378



Weighted Moving Average

- The simple moving average formula implies equal weighting for all periods (簡單移動平均假設不同時間的資料權重均相同).
- A weighted moving average allows unequal weighting of prior time periods.
  - The sum of the weights must be equal to one.
  - Often, more recent periods are given higher weights than periods farther in the past.

$$F_t = w_1 A_{t-1} + w_2 A_{t-2} + \cdots + w_n A_{t-n}$$

  - 加權移動平均的問題在於不知如何決定不同時間點的資料權重

Exponential Smoothing

- A weighted average method that includes all past data in the forecasting calculation
- More recent results weighted more heavily
- The most used of all forecasting techniques
- An integral part of computerized forecasting
- Well accepted for six reasons
  - ✓ Exponential models are surprisingly accurate.
  - ✓ Formulating an exponential model is relatively easy.
  - ✓ Little computation is required to use the model.
  - ✓ Computer storage requirements are small.
  - ✓ Tests for accuracy are easy to compute.

Exponential Smoothing Model

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1}) = \alpha A_{t-1} + (1-\alpha) F_{t-1}$$

$$= \alpha A_{t-1} + (1-\alpha)[\alpha A_{t-2} + (1-\alpha) F_{t-2}] \ldots\ldots$$

where

Alpha一般介於0.1~0.5之間

F_t = The exponentially smoothed forecast for period t

F_{t-1} = The exponentially smoothed forecast made for the prior period

A_{t-1} = The actual demand in the prior period

α = The desired response rate, or smoothing constant

指數平滑法假設越近的資料越重要，越遠的資料越不重要

Exponential Smoothing Example

Week	Demand	Forecast
1	820	820
2	775	820
3	680	811
4	655	785
5	750	759
6	802	757
7	798	766
8	689	772
9	775	756
10		760

$F_2 = F_1 + \alpha(A_1 - F_1) = 820 + 0.2(820-820)$

$F_4 = F_3 + \alpha(A_3 - F_3) = 811 + 0.2(680-811)$

$F_6 = F_5 + \alpha(A_5 - F_5) = 759 + 0.2(750-759)$

$F_8 = F_7 + \alpha(A_7 - F_7) = 766 + 0.2(798-766)$

$F_{10} = F_9 + \alpha(A_9 - F_9) = 756 + 0.2(775-756)$

Criteria for Selecting Methods

- Criteria to use in making forecast method and parameter choices include
  1. Minimizing MAPE, MAD, or MSE
  2. Meeting managerial expectations of changes in the components of demand
  3. Minimizing the forecast error last period

- Statistical performance measures can be used
  1. For projections of more stable demand patterns, use lower α values or larger n values
  2. For projections of more dynamic demand patterns try higher α values or smaller n values

Holt-Winters moving average

$$L_t = \alpha(y_t - S_{t-s}) + (1-\alpha)(L_{t-1} + b_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

$$S_t = \gamma(y_t - L_t) + (1-\gamma)S_{t-s}$$

$$F_{t+k} = L_t + kb_t + S_{t+k-s}$$

Forecast consists of level (L), trend (T), and seasonal (S) components

MLR的判定係數

傳統上， MLR (multiple linear regression)迴歸模型的評估常使用判定係數(coefficient of determination): $Y=a_0+a_1x_1+a_2x_2+a_3x_3+\ldots$

$$R^2 = SSR/SST$$

或調整後判定係數（adjusted coefficient of determination:）避免使用過多沒必要的X (k表示X的個數, T表示樣本數)

$$\bar{R}^2 = 1-\frac{T-1}{T-k}(1-R^2)$$

MARS介紹

Multivariate Adaptive Regression Splines
(MARS多元自適雲形迴歸)

統計物理學家 Friedman 於 1991 年提出多元適應雲形迴歸，其屬於一種多變量無母數迴歸處理的程序。其能夠正確的預測連續與二元的相依變數，找出隱藏在高維度資料中的複雜結構及變數的交互作用。

過去對於資料之分析，常常不論資料是否符合常態，均用線性迴歸進行實證，導致模型適用性出現問題，而MARS為多變量無母數分析技術，相較於傳統之線性迴歸，可解決資料非線性、非常態等問題。

MARS原理

MARS 基本概念為運用分段基底函數(Spline Basis Function, BF)加總，以組合較有彈性的預測模型解決複雜的資料配適。

$$\hat{f}(x) = a_0 + \sum_{m=1}^{M} a_m \prod_{k=1}^{K_m} [s_{km} \cdot (x_{v(k,m)} - t_{km})]_+$$

有關BF之後段累程之部分，主要是根據需求變化而得：

$$B_m(x) = \prod_{k=1}^{K_m} H[s_{km} \cdot (x_{v(k,m)} - t_{km})]_+$$

上式中a_0與a_m皆為參數值；M為BF的個數； K_m為切割之節點(knot)個數； S_{km}之值為+1或-1，作用為顯示方向；v(k,m)是對變數的標示；t_{km}是各節點的臨界值。

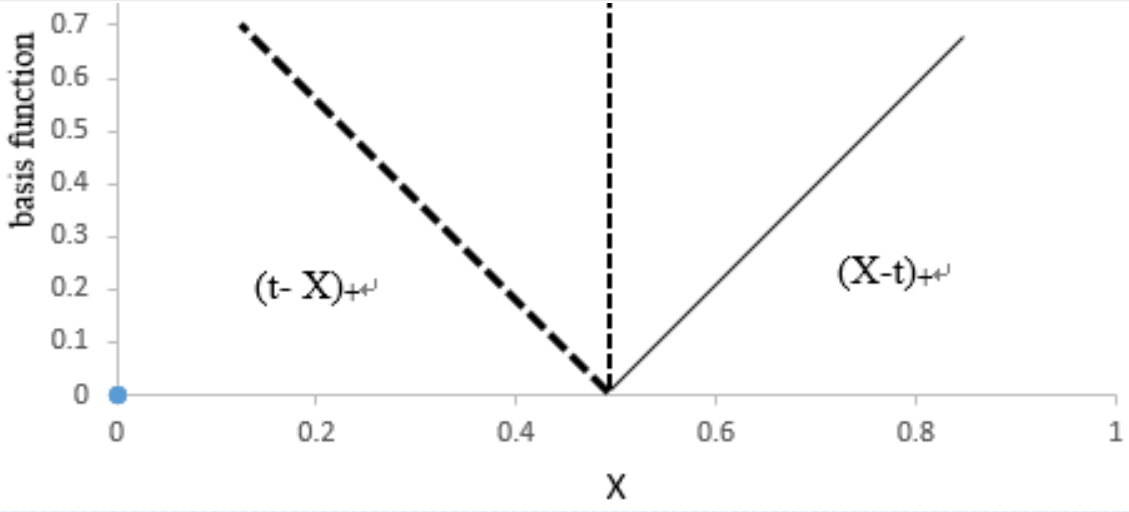
MARS基底函數

- 模式中所以用的片段線性基底函數形式如下(+表示正值)：

$(X-t)_+ = \{X-t, \text{ if } X > t \text{ ; } 0, \text{ otherwise}\}$，

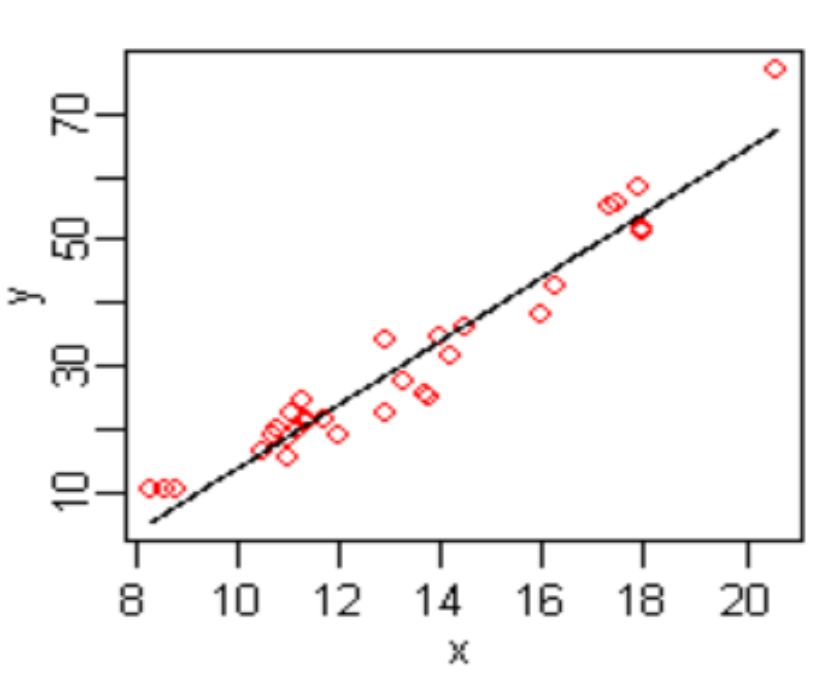
$(t-X)_+ = \{t-X, \text{ if } X < t \text{ ; } 0, \text{ otherwise}\}$

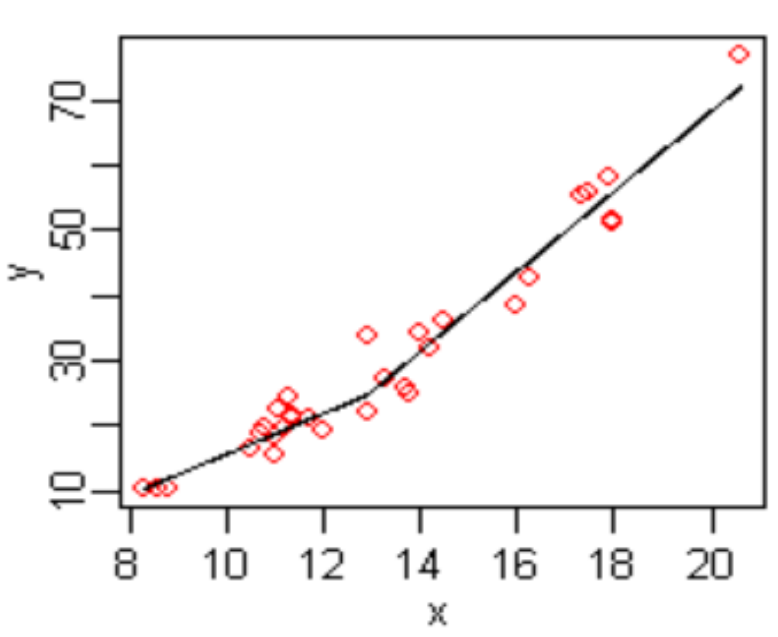
- X為預測變數，t為獨立的觀察值，表示折點(knot)的位置；通常$(X-t)_+$與$(t-X)_+$會成對出現。例如：t=0.5時，基底函數為$(X-0.5)_+$與$(0.5-X)_+$，如圖二所示。



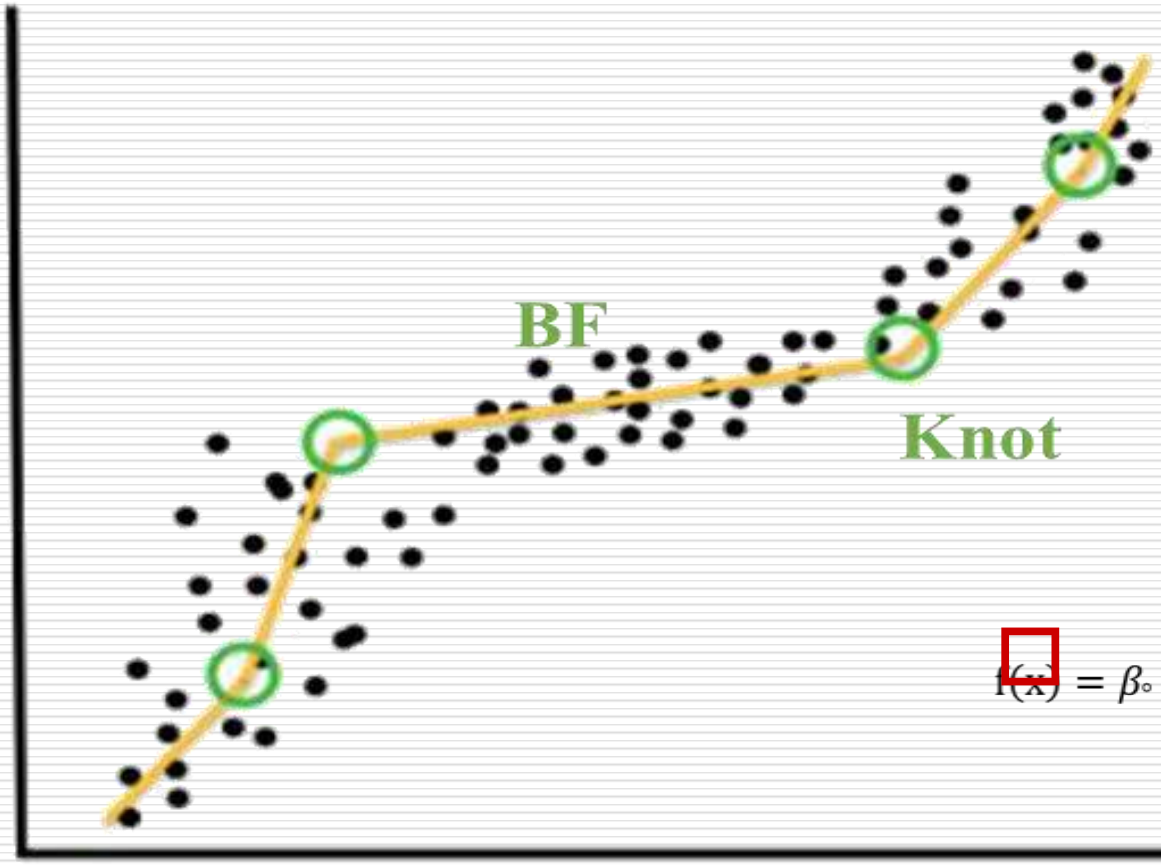
片斷線性基底函數，只會依據測試，選取左邊或右邊的任一片段當作基底的一員

MARS與MLR的比較





MARS使用片段線性基底進行估計



$$f(x) = \beta_\circ + \sum_{m=1}^{M} \beta_m h_m(X)$$

MARS執行過程

篩選的過程採用前推式與後推式(Forward and Backward Stepwise Procedure) **的演算法**：

(一)前推式階段

不斷增加解釋方程式(**所有的主要效果、新的主要影響因子、折點或交互作用**)**的個**數，使模**型**不斷成長直到找出一個完整模型為止。

注意：在此階段會有過度配適(over fitting)的問題



MARS執行過程

(二)後推式階段 (general cross validation)

此階段根據缺適性的概念加以判斷，選擇重要變數的過程中使用 GCV 法為判定原則，所以當一變數由模型中去除後使 GCV 顯著降低，則可知此變數為重要關鍵要素。

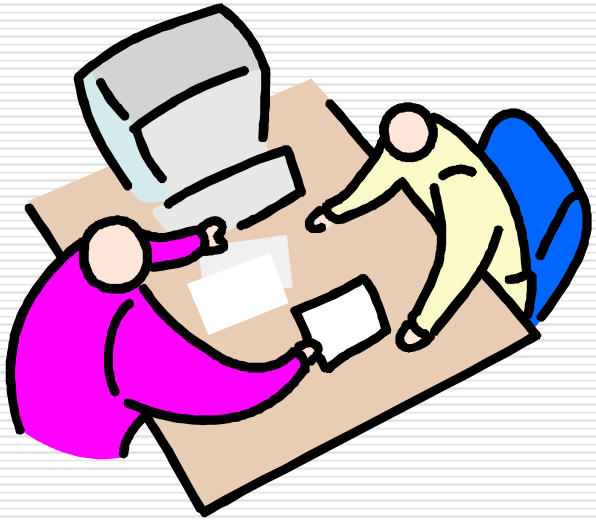
$$GCV(M) = \frac{1}{N}\sum_{i=1}^{N}\frac{\left[y_i - \hat{f}_M(\pi_i)\right]^2}{\left[1 - \frac{C(M)}{N}\right]^2}$$

上式中分子為實際值減預測值，
C(M)為採用M個BF所花費的成本。



因應價格管制的措施

- 調整產品組合
- 改變目標市場與區隔
- 上市新產品或修改現有產品的形式
- 和政府溝通或談判
- 預先採取因應措施



傾銷的定義

- 進口商品以低於母國或來源國普遍的價位來定價。
- 如果其價格是低於該產品的生產成本再加上廠商應得的8%利潤，或是低於生產國的普遍價格，則可判定其為傾銷。



傾銷的類別

- 掠奪性傾銷（Predatory Dumping）：指某一跨國公司以增加國內的生產成本為代價，或是刻意虧損，來達成在另一個國家低價銷售以奪取市場占有率的目的。
- 無心傾銷（Unintentional Dumping）：由於交易日、運送日，和到達日之間的時間落差而產生的價格差異。由於價格隱含著匯率的因素，當匯率因為時間的關係而改變時，則有可能造成最終售價低於其生產成本或母國的當地價格，因此在無意間造成傾銷。

因應競爭廠商傾銷的措施

- 產品升級
- 強化服務
- 配銷與溝通
- 採取訴訟



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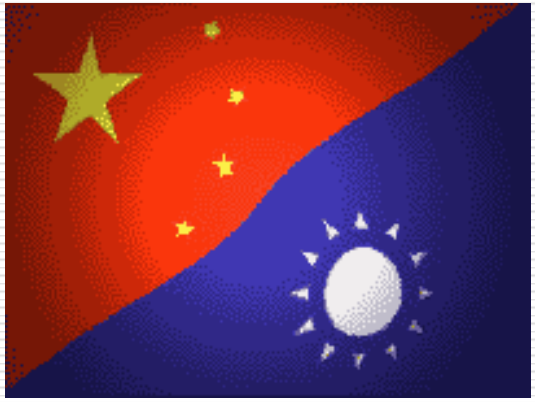
因應價格攀升的措施

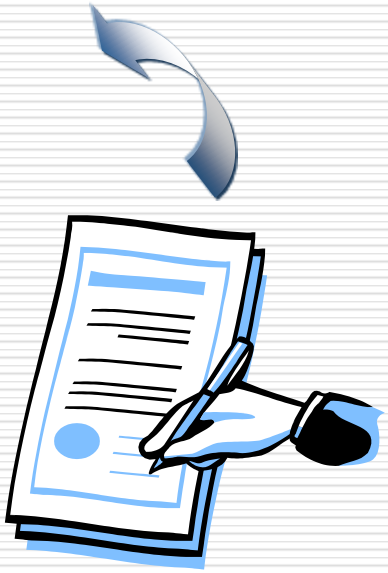
- 重新安排通路
- 刪減某些產品屬性
- 在國外生產或組裝產品
- 修改產品規格
- 重新定位



因應高通貨膨脹的措施

- 修改產品
- 調整原物料供應廠商
- 降低投資成本
- 透過長期合約來避險
- 以相對較穩定的貨幣來報價
- 追求快速的存貨周轉率
- 向曾在高通貨膨脹國家運作的廠商學習





Understanding Inflation

Inflation: Increase in amount of money needed to purchase *same amount of goods or services.* Inflation results in a *decrease in purchasing power, i.e., one unit of money buys less goods or services*

Two ways to work problems *when considering inflation:*

(1) Convert to constant value (CV) dollars, then use real rate *i*.
If *f* = inflation rate (% per year), the equation is:

$$\text{Constant-value dollars} = \frac{\text{future dollars}}{(1+f)^n}$$

(2) Leave money amounts ***as is*** and use ***interest rate adjusted for inflation, i_f***

$$i_f = i + f + (i)(f)$$

Example: Constant Value Dollars

How much would be *required today* to purchase an item that increased in cost by exactly the inflation rate? The cost 30 years ago was $1000 and inflation has consistently averaged 4% per year.

Solution: Solve for future dollars

$$\text{Future dollars} = \text{constant value dollars}(1 + f)^n$$
$$= 1000(1 + 0.04)^{30}$$
$$= \$3243$$

Note: This calculation only accounts for the *decreased purchasing power of the currency*. It does ***not*** take into account the ***time value of money*** (to be discussed)

Deflation: Opposite of inflation; purchasing power of money is *greater* in future than at present; however, money, credit, jobs are 'tighter'

Three Different Rates

- Real or inflation rate i – Rate at which interest is earned when *effects of inflation are removed*; i represents the real increase in purchasing power

- Market or inflation-adjusted rate i_f – Rate that *takes inflation into account.* Commonly stated rate everyday

$$P = F\frac{1}{(1+i_f)^n} = F(P/F,i_f,n)$$

- Inflation rate f – Rate of *change in value of currency*

Relation between three rates is derived using the relation

Market rate is: $i_f = i + f + (i)(f)$

Example: Market vs. Real Rate

Money in a medium-risk investment makes a guaranteed 8% per year. Inflation rate has averaged 5.5% per year. What is the real rate of return on the investment?

Solution: Solve for the real rate i in relation for i_f

$$i_f = i + f + (i)(f)$$

$$i = \frac{1 + f}{i_f - f}$$

$$= \frac{0.08 - 0.055}{1 + 0.055}$$

$$= 0.024$$

Investment pays only 2.4% per year in real terms vs. the stated 8%

PW Calculations with Inflation

Two ways to account for inflation in PW calculations

(1) Convert cash flow into *constant-value* (CV) *dollars* and use regular i

where: CV = future dollars/$(1 + f)^n$ = then-current dollars/$(1 + f)^n$

f = inflation rate

(**Note:** *Calculations up to now have assumed constant-value dollars*)

(2) Express cash flow in *future (then-current) dollars* and use inflated interest rate where $i_f = i + f + (i)(f)$

(**Note:** *Inflated interest rate is the market interest rate*)

Example: PW with Inflation

A honing machine will have a cost of $25,000 (future cost) six years from now. Find the PW of the machine, if the real interest rate is 10% per year and the inflation rate is 5% per year using (a) constant-value dollars, and (b) future dollars.

Solution: (a Determine *constant-value* dollars and *use i* in PW equation

$$\mathbf{CV} = 25{,}000 / (1 + 0.05)^6 = \$18{,}655$$

$$\mathbf{PW} = 18{,}655(P/F,10\%,6)$$
$$= \$10{,}530$$

(b) Leave as future dollars and *use* i_f in PW equation

$$i_f = 0.10 + 0.05 + (0.10)(0.05) = 15.5\%$$

$$\mathbf{PW} = 25{,}000(P/F,15.5\%,6)$$
$$= \$10{,}530$$

FW Calculations with Inflation

FW values can have *four different* interpretations

(1) The *actual amount accumulated*

✓ Use i_f in FW equation → **FW = PW(F/P, i_f, n)**

(2) The *purchasing power* in terms of CV dollars *of the future amount*

✓ Use i_f in FW equation and divide by $(1+f)^n$ or **use real *i***

where real $i = (i_f - f)/(1 + f)$ → **FW = PW(F/P,i,n)**

(3) The *number of future dollars required to have the same purchasing power* as a dollar today with no time value of money considered

✓ Use *f* instead of *i* in F/P factor → **FW = PW(F/P,f,n)**

(4) The amount required to *maintain the purchasing power of the present sum **and** earn a stated real rate of return*

✓ Use i_f in FW equation → **FW = PW(F/P, i_f, n)**

Example: FW with Inflation

An engineer invests $15,000 in a savings account that pays interest at a real 8% per year. If the inflation rate is 5% per year, Determine (a) the amount of money that will be accumulated in 10 years, (b) the purchasing power of the accumulated amount (in terms of today's dollars), (c) the number of future dollars that will have the same purchasing power as the $15,000 today, and (d) the amount to maintain purchasing power and earn a real 8% per year return.

Solution:

(a)The *amount accumulated* is a function of the *market interest rate*, i_f

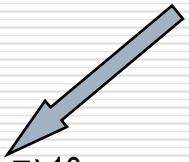
$$i_f = 0.08 + 0.05 + (0.08)(0.05) = 13.4\%$$

Amount Accumulated = 15,000(F/P,13.4%,10)

= **$52,750**

Example: FW with Inflation (cont'd)

(b) To find the *purchasing power* of the accumulated amount *deflate* the inflated dollars



Purchasing power = 15,000(F/P,13.4%,10) / $(1 + 0.05)^{10}$
=15,000(F/P, 8%,10) = **$32,384**

(c) The number of future dollars required to purchase goods that cost $15,000 now is the inflated cost of the goods

Number of future dollars = 15,000(F/P,5%,10)
= **$24,434**

(d) In order to maintain purchasing power *and* earn a real return, money must grow by the inflation rate ***and*** the interest rate, or i_f = 13.4%, as in part (a)

FW = 15,000(F/P,13.4%,10)
= **$52,750**